

Day 83 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in TN History/Culture/Heritage (5): The Music Academy's Centenary Awards 2026

READING MATERIAL

Covers The Music Academy, Madras's prestigious annual Carnatic music and dance titles — awarded in 2026 to mark the Academy's centenary.

Q: What is the Sangita Kalanidhi award, and who bestows it?

A: The highest honour in Carnatic music (often called the 'Nobel Prize' of the field), awarded annually for lifetime contributions to music — bestowed by The Music Academy, Madras (founded 1928), which is celebrating its centenary in 2026-27.

Q: Who received the Sangita Kalanidhi award in 2026, and for what is she known?

A: Jayanthi Kumaresh — a veena exponent and sixth-generation musician, known for 'globalising the Saraswati Veena', founding the Indian National Orchestra, and performing at the United Nations and the Théâtre de la Ville in Paris.

Q: Who received the Nritya Kalanidhi (dance) title in 2026, and what milestone did it mark?

A: Malavika Sarukkai, a veteran Bharatanatyam dancer — the title specifically marked the Music Academy's 100th (centenary) Dance Festival.

MOCK TEST (WITH ANSWERS)

1. The Sangita Kalanidhi award is often described as the 'Nobel Prize' of:

- A. Bharatanatyam
- B. Carnatic music ✓**
- C. Tamil literature
- D. Classical painting

Explanation: Carnatic music. [easy, web-research]

2. The Sangita Kalanidhi and Nritya Kalanidhi titles are bestowed by:

- A. Sangeet Natak Akademi
- B. The Music Academy, Madras ✓**
- C. Tamil Nadu Government
- D. Sahitya Akademi

Explanation: The Music Academy, Madras (founded 1928). [medium, web-research]

3. Jayanthi Kumaresh, the Sangita Kalanidhi 2026 recipient, is known as an exponent of which instrument?

- A. Mridangam
- B. Veena ✓**
- C. Violin
- D. Flute

Explanation: Veena (Saraswati Veena). [medium, web-research]

4. Malavika Sarukkai received the Nritya Kalanidhi 2026 title, marking the Music Academy's:

- A. 50th Dance Festival
- B. 75th Dance Festival
- C. 100th Dance Festival ✓**
- D. 125th Dance Festival

Explanation: 100th (centenary) Dance Festival. [medium, web-research]

5. The Music Academy, Madras was founded in:

- A. 1916
- B. 1928 ✓**
- C. 1947

D. 1956

Explanation: 1928. [hard, web-research]

Part B – Aptitude & Mental Ability: Highest Common Factor (HCF)

READING MATERIAL

HCF is 1 of 10 named Aptitude sub-skills. Real exam evidence (4 confirmed hits across 2022/2024/2025) shows TNPSC favors HCF *application* problems — measuring/tiling/container word problems and the HCF-LCM relationship — over plain 'find the HCF of these two numbers' questions, so the reading below leads with those application shapes.

Q: Real exam (2025): Two numbers are in the ratio 15:11. If their HCF is 13, find the numbers.

A: When two numbers are in a given ratio, write them as $15k$ and $11k$. Since $\text{HCF}(15,11)=1$ (the ratio is already in lowest terms), k IS the HCF — so $k=13$. The numbers are $15 \times 13=195$ and $11 \times 13=143$.

Q: Real exam (2025): What is the least number of square tiles needed to pave the ceiling of a room 15m 17cm long and 9m 2cm broad?

A: Convert to one unit: $1517 \text{ cm} \times 902 \text{ cm}$. The LARGEST square tile that exactly fits (no cutting) has a side equal to the HCF of the two dimensions. $\text{HCF}(1517, 902) = 41 \text{ cm}$ (via the Euclidean algorithm: $1517=1 \times 902+615$; $902=1 \times 615+287$; $615=2 \times 287+41$; $287=7 \times 41+0$). Tile area = $41 \times 41 = 1,681 \text{ cm}^2$. Room area = $1517 \times 902 = 1,368,334 \text{ cm}^2$. Number of tiles = $1,368,334 \div 1,681 = 814$.

Q: Real exam (2024): If the HCF of x^3-8y^3 and $x^2-k^2y^2$ is $x+ky$, find k .

A: Factor x^3-8y^3 using $a^3-b^3=(a-b)(a^2+ab+b^2)$ with $a=x$, $b=2y$: $x^3-8y^3 = (x-2y)(x^2+2xy+4y^2)$. Factor $x^2-k^2y^2$ using $a^2-b^2=(a-b)(a+b)$: $x^2-k^2y^2 = (x-ky)(x+ky)$. For ' $x+ky$ ' to be the common factor (HCF) of both, it must match one of the first expression's factors. $(x-2y)$ is the only linear factor available there, so $x+ky$ must equal $x-2y$, giving $k=-2$.

Q: Real exam (2022): What is the greatest possible volume of a vessel that can exactly measure the milk in cans of 80, 100, and 120 litres (each can measured to full capacity only)?

A: This is asking for the HCF of 80, 100, and 120. $\text{HCF}(80,100)=20$, then $\text{HCF}(20,120)=20$. So the answer is 20 litres — the largest vessel that can measure each can's full volume an exact whole number of times.

Q: Real exam (2022): The HCF and LCM of two numbers are 16 and 240 respectively. If one number is 48, find the other.

A: Use the relationship: $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$. $16 \times 240 = 3,840$. Other number = $3,840 \div 48 = 80$.

Q: Method — finding HCF via the Euclidean algorithm (the fast way for large numbers): What is the HCF of 1517 and 902?

A: Repeatedly divide the larger by the smaller and replace the larger with the remainder, until the remainder is 0 — the last nonzero remainder is the HCF. $1517=1 \times 902+615 \rightarrow 902=1 \times 615+287 \rightarrow 615=2 \times 287+41 \rightarrow 287=7 \times 41+0$. $\text{HCF}=41$. This is the same calculation used in ro2 above — recognize it as the standard fast method rather than listing all factors of both numbers.

MOCK TEST (WITH ANSWERS)

1. Two numbers are in the ratio 21:14. If their HCF is 12, find the larger number.

A. 36 ✓

B. 24

C. 252

D. 168

Explanation: Watch out: 21 and 14 are NOT coprime ($\text{gcd}=7$), so you can't directly multiply them by the HCF. Reduce the ratio first: $21:14 = 3:2$ (coprime). Now apply the HCF: larger = $3 \times 12=36$, smaller = $2 \times 12=24$. [hard, authored]

2. Find the largest square tile that can exactly pave a floor measuring 3m 8cm by 4m 20cm (no cutting).

A. 28 cm ✓

B. 14 cm

C. 56 cm

D. 84 cm

Explanation: Convert to cm: 308 and 420. $\text{HCF}(308,420)$: $420=1 \times 308+112$; $308=2 \times 112+84$; $112=1 \times 84+28$; $84=3 \times 28+0$. $\text{HCF}=28 \text{ cm}$. [hard, authored]

3. If the HCF of a^2-4b^2 and $a^2+ab-6b^2$ is $a-2b$, this is consistent with which factorization of the second expression?

A. $(a+3b)(a-2b)$ ✓

B. $(a-3b)(a+2b)$

C. $(a+3b)(a+2b)$

D. $(a-3b)(a-2b)$

Explanation: $a^2-4b^2=(a-2b)(a+2b)$. The second expression factors as $(a+3b)(a-2b) = a^2+ab-6b^2$ — sharing the common factor $(a-2b)$, so that's the HCF. [hard, authored]

4. What is the greatest volume that can exactly measure the contents of drums holding 72, 96, and 120 litres?

A. 12 litres

B. 24 litres ✓

C. 8 litres

D. 6 litres

Explanation: $\text{HCF}(72, 96) = 24$, then $\text{HCF}(24, 120) = 24$. Answer: 24 litres. [medium, authored]

5. The HCF and LCM of two numbers are 12 and 180 respectively. If one number is 36, find the other.

A. 48

B. 54

C. 60 ✓

D. 72

Explanation: $\text{HCF} \times \text{LCM} = 12 \times 180 = 2,160$. Other number = $2,160 \div 36 = 60$. [medium, authored]

6. Using the Euclidean algorithm, find the HCF of 1,073 and 925.

A. 37 ✓

B. 29

C. 31

D. 43

Explanation: $1073 = 1 \times 925 + 148$; $925 = 6 \times 148 + 37$; $148 = 4 \times 37 + 0$. $\text{HCF} = 37$. [medium, authored]

7. Two numbers are in the ratio 8:9 and their HCF is 7. Find their sum.

A. 105

B. 119 ✓

C. 126

D. 112

Explanation: Numbers: $8 \times 7 = 56$ and $9 \times 7 = 63$. Sum = $56 + 63 = 119$. [medium, authored]

Part C – General English: Vocabulary: Idioms, Phrases, and Phrasal Verbs (Expansion)

READING MATERIAL

More idioms, fixed phrases, and phrasal verbs beyond the first Idioms/Phrasal-Verbs topic — no overlap with previously shipped vocabulary content.

Q: What do the idioms 'Find the pony' and 'Kettle of fish' mean?

A: Find the pony = to find something good in a bad situation. Kettle of fish = a difficult or messy situation.

Q: What do the idioms 'Sleep like a log' and 'Upset the apple cart' mean?

A: Sleep like a log = to sleep very deeply. Upset the apple cart = to cause trouble or disrupt a plan.

Q: What do the phrasal verbs 'look out', 'look into', 'look up to', and 'look up' mean?

A: Look out = be careful. Look into = investigate. Look up to = admire (someone). Look up = search for (information).

Q: What is a synonym for 'Affluent' (as in 'He was the spoilt child of Affluent parents')?

A: Wealthy (not poor, intelligent, or useless).

Q: What do the phrasal verbs 'bring up' and 'call off' mean?

A: Bring up = raise a topic. Call off = cancel.

Q: What do the phrasal verbs 'give up' and 'carry out' mean?

A: Give up = quit. Carry out = execute/perform.

Q: What does the phrasal verb 'turn down' mean?

A: Reject (e.g., 'turn down an offer') — not 'reduce volume', which is a different, less common sense of the same phrase.

MOCK TEST (WITH ANSWERS)

1. 'Kettle of fish' means:

A. A tasty meal

B. A difficult or messy situation ✓

C. A type of seafood dish

D. A friendly gathering

Explanation: A difficult or messy situation. [medium, question-bank-2025]

2. 'Upset the apple cart' means:

A. To cause trouble or disrupt a plan ✓

B. To eat too many apples

C. To win unexpectedly

D. To clean up a mess

Explanation: To cause trouble or disrupt a plan. [medium, question-bank-2025]

3. Choose the correct meaning of the phrasal verb 'bring up'.

A. Raise a topic ✓

B. Throw out

C. Cancel

D. Shout loudly

Explanation: Raise a topic. [easy, web-research]

4. Choose the correct meaning of the phrasal verb 'call off'.

A. Postpone

B. Cancel ✓

C. Invite

D. Attend

Explanation: Cancel. [easy, web-research]

5. Choose the correct meaning of the phrasal verb 'give up'.

A. Start

B. Quit ✓

C. Increase

D. Offer again

Explanation: Quit. [easy, web-research]

6. Choose the correct meaning of the phrasal verb 'carry out'.

A. Cancel

B. Ignore

C. Execute ✓

D. Forget

Explanation: Execute. [medium, web-research]

7. Choose the correct meaning of the phrasal verb 'turn down'.

A. Reduce volume

B. Reject ✓

C. Sit down

D. Fall

Explanation: Reject. [medium, web-research]

8. 'Look up to' someone means:

A. To search for them

B. To admire them ✓

C. To avoid them

D. To investigate them

Explanation: To admire them. [medium, question-bank-2025]

9. A synonym for 'Affluent' is:

A. Poor

B. Intelligent

C. Wealthy ✓

D. Useless

Explanation: Wealthy. [easy, question-bank-2022]